

Monte Carlo uncertainty on a predicted lattice thermal conductivity

The method used in this project, written out

Companion to `notebooks/monte_carlo_explained.ipynb`, which implements every equation below and verifies it reproduces `scripts/cgcnn/07_predict_kappa.py` exactly.

1 The problem

The pipeline predicts two elastic moduli and turns them into a lattice thermal conductivity through a chain of closed-form relations:

$$v_l = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}, \quad v_t = \sqrt{\frac{G}{\rho}}, \quad v_s = \left[\frac{1}{3} \left(\frac{1}{v_l^3} + \frac{2}{v_t^3} \right) \right]^{-1/3} \quad (1)$$

$$\nu = \frac{(v_l/v_t)^2 - 2}{2(v_l/v_t)^2 - 2}, \quad \gamma = \frac{3(1 + \nu)}{2(2 - 3\nu)} \quad (2)$$

$$\kappa_L = \frac{G v_s V^{1/3}}{N T} e^{-\gamma} \quad (3)$$

Every one of these is exact. But K and G are *predictions*, and a prediction has an uncertainty. The question is what that uncertainty does to κ_L .

1.1 Why the textbook answer does not work here

The standard tool is the *delta method*: linearise about the prediction and combine partial derivatives,

$$\sigma_\kappa^2 \approx \left(\frac{\partial \kappa_L}{\partial K} \right)^2 \sigma_K^2 + \left(\frac{\partial \kappa_L}{\partial G} \right)^2 \sigma_G^2.$$

This is a first-order Taylor expansion, and it is only as good as the assumption that κ_L is close to linear over the width of the uncertainty. Ours is not. Reading the chain from the inside out: K and G enter v_l/v_t as a ratio under square roots; that ratio enters ν through a rational function; ν enters γ through another rational function with $(2 - 3\nu)$ in the denominator; and γ then enters *exponentially*. Composing four non-linear steps and expecting a linear response over a $\pm 15\%$ input range is not reasonable, and the derivatives themselves are unpleasant to write down correctly.

The Monte Carlo alternative

Instead of differentiating the function, evaluate it many times on inputs drawn to represent the uncertainty, and look at the spread of the outputs. No derivatives, no linearity assumption, and the answer becomes exact as the number of draws grows. The cost is arithmetic, which is cheap.

2 What the uncertainty is, and where it comes from

The moduli come from an **ensemble**: three networks trained from different random initialisations on the same data and split. For crystal i they produce three values, whose mean is the

prediction and whose standard deviation is the uncertainty. Both are computed on $\log_{10}$ of the modulus, because that is the space the networks are trained in:

$$\mu_{K,i} = \log_{10} \hat{K}_i, \quad \sigma_{K,i} = \text{sd}(\log_{10} K_i^{(1)}, \log_{10} K_i^{(2)}, \log_{10} K_i^{(3)}),$$

and likewise for G . In this project's 1,213-crystal prediction set the median σ_K is 0.0235 and the median σ_G is 0.0295, i.e. roughly 5–7% multiplicative uncertainty.

3 The method

3.1 Step 1 — draw

For each crystal i and each sample $j = 1 \dots M$,

$$\log_{10} K_i^{(j)} = \mu_{K,i} + \sigma_{K,i} z_{K,i}^{(j)}, \quad z_K \sim \mathcal{N}(0, 1) \quad (4)$$

$$\log_{10} G_i^{(j)} = \mu_{G,i} + \sigma_{G,i} z_{G,i}^{(j)}, \quad z_G \sim \mathcal{N}(0, 1) \quad (5)$$

then $K_i^{(j)} = 10^{\log_{10} K_i^{(j)}}$ and similarly for G . This is the location–scale construction: a standard normal scaled by σ and shifted by μ is distributed $\mathcal{N}(\mu, \sigma^2)$.

Two modelling choices are embedded here and both deserve to be stated rather than assumed.

Why a normal distribution. Three members give a mean and a spread and essentially no information about shape. Among all distributions with a given mean and variance, the normal is the maximum-entropy choice — the one that adds the least unwarranted structure.

Why z_K and z_G are independent. The K ensemble and the G ensemble are separately seeded runs sharing no weights and no random state, so nothing links their draws. *This is the weakest assumption in the method.* Both ensembles see the same crystal, and a crystal that is hard for one is plausibly hard for the other; a positive correlation between the two errors would widen the true interval beyond what this produces.

3.2 Step 2 — propagate

Every sampled pair goes through the *same* function used for the point estimate:

$$\kappa_{L,i}^{(j)} = f(K_i^{(j)}, G_i^{(j)}; V_i, \rho_i, \bar{M}_i, N_i),$$

where f is equations (1)–(3) and the structural quantities are held **fixed**: they are read from the CIF, not predicted, so they carry no uncertainty of this kind.

Because the point estimate and the interval come from the same f , they are consistent by construction rather than by a separate approximation — the median of the samples cannot drift away from the reported central value for reasons of method.

3.3 Step 3 — summarise

With M values of κ_L per crystal, report the empirical quantiles

$$p_{05,i} = Q_{0.05}(\{\kappa_{L,i}^{(j)}\}), \quad p_{50,i} = Q_{0.50}(\cdot), \quad p_{95,i} = Q_{0.95}(\cdot).$$

Why quantiles, and not mean $\pm 2\sigma$

The input distribution is symmetric; the output is not. On the worked example in the notebook (LaGa) the median sits at $4.48 \text{ W m}^{-1} \text{K}^{-1}$, with 2.31 down to p_{05} and 4.68 up to p_{95} — the upper half of the interval is twice the lower. A symmetric interval would misstate the answer in both directions. Quantiles assume nothing about shape.

4 Why everything happens in $\log_{10}$

This is the design choice that is easiest to overlook and most consequential.

The error is multiplicative. The networks are trained on $\log_{10}(\text{modulus})$ against a log-space loss, so their error is naturally a percentage, not a number of GPa. An error of 12 GPa is catastrophic on a 20 GPa crystal and negligible on a 400 GPa one; an error of 15% means the same thing on both. Sampling in log space encodes that correctly.

Negative moduli become impossible. Since $10^x > 0$ for every real x , a log-space draw can never be non-positive. Sampling additively in GPa can, and does: on this project’s softest crystal (Rb, $\hat{K} = 2.01$ GPa, $\sigma_K = 0.1285$), an equivalent-width additive draw of 20,000 samples produced 5 negative moduli. A negative modulus is unphysical, and it makes $\sqrt{G/\rho}$ undefined — the Monte Carlo would fail on exactly the soft crystals the screen exists to find.

5 How many samples

Monte Carlo error falls as $M^{-1/2}$, so accuracy is cheap but improves slowly. This project uses $M = 2000$. Measured convergence on the worked crystal, against a 20,000-sample reference:

M	p_{05}	p_{95}
50	2.466	8.517
250	2.039	9.199
1,000	2.234	9.316
2,000	2.172	9.160
5,000	2.155	9.356
20,000	2.169	9.339

At $M = 2000$, p_{05} is within 0.13% and p_{95} within 1.92% of the reference. Both are far smaller than the interval itself, which spans a factor of 4.2. Increasing M further would refine a number whose real uncertainty is dominated by something else entirely — which is the subject of the next section.

6 The interval is too narrow, and by how much

Everything above is arithmetically correct and still produces an interval that is badly overconfident. This is the most important result in this document.

Step 25 of the pipeline tested it on the held-out set, where the true DFT moduli are known, and asked: of the crystals where the method claimed $X\%$ confidence, what fraction did the interval actually contain?

nominal	K_{VRH}		G_{VRH}	
	raw	recalibrated	raw	recalibrated
0.5	0.234	0.496	0.214	0.512
0.8	0.414	0.789	0.394	0.811
0.9	0.508	0.895	0.479	0.909

A nominal 90% interval covers about **50%** of true values.

6.1 Why

The Monte Carlo faithfully propagates the ensemble’s *disagreement*. But disagreement is only the variance component of the error:

$$\mathbb{E}[(\hat{y} - y)^2] = \underbrace{(\mathbb{E}[\hat{y}] - y)^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{y})}_{\text{what the ensemble sees}}.$$

Three networks with the same architecture, the same training data and the same split share their blind spots. When all three are wrong in the same direction, the bias term is large and the variance term is small — the ensemble agrees with itself, the sampled spread is narrow, and the method reports high confidence in a wrong answer. Measured here, member disagreement runs 4–6× smaller than the true error.

The generalisable lesson

An ensemble’s spread is a lower bound on its uncertainty, never the whole of it. It must be calibrated against held-out truth before it is quoted as a confidence interval. Rescaling σ by a factor fitted on validation data restores $\sim 90\%$ coverage out of sample (right-hand columns above) — but the factor has to be measured, and it is level-dependent: roughly 4.2× at nominal 0.8 and 7.0× at 0.9. Applying that correction to this project’s DFT shortlist collapsed almost every “tight” confidence label to “wide”.

7 Summary

step	what happens	why
draw	$M = 2000$ pairs from $\mathcal{N}(\mu, \sigma^2)$ in $\log_{10}$ space	the error is multiplicative, and 10^x cannot be negative
propagate	every pair through the same f as the point estimate	no linearisation, no derivatives; interval consistent with the central value by construction
summarise	the 5th, 50th and 95th percentiles	the output is skewed although the input was symmetric

And the caveat that outweighs the rest: this propagates *variance* only. Uncorrected, the interval covers about half of what it claims.